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This study identified cognitive processes that underlie individual differences in working memory (WM) and mathematical problem-solution accuracy in elementary school children at risk and not at risk for serious math difficulties (SMD). A battery of tests was administered that assessed problem solving, achievement, and cognitive processing in children in first ($N = 130$), second ($N = 92$) and third grades ($N = 131$). The results were that (a) younger children and children at risk for SMD performed poorer on WM and problem-solving tasks, as well as measures of math calculation, reading, semantic processing, phonological processing, and inhibition, than older children and children not at risk for SMD and (b) WM predicted solution accuracy of word problems independent of measures of fluid intelligence, reading skill, math skill, knowledge of algorithms, phonological processing, semantic processing, speed, short-term memory, and inhibition. The results support the notion that the executive system is an important predictor of children's problem solving.

Word problems constitute one of the most important mediums through which students can potentially learn to select and apply strategies necessary for coping with everyday problems. To comprehend and solve mathematical word, or story, problems, one must be able to keep track of incoming information (see, e.g., Anderson, Reder, & Lebiere, 1996; Kail & Hall, 1999; Mayer & Hegarty, 1996; Swanson & Sachse-Lee, 2001). This is necessary to understand words, phrases, sentences, and propositions that, in turn, are necessary to construct a coherent and meaningful interpretation of word problems. Temporary storage of material that has been read or heard is said to depend on working memory (WM; see, e.g., Baddeley & Logie, 1999; Case, 1995), which takes into account the storage of items for later retrieval and which is a function of the individual's level of text processing (see, e.g., Engle, Tuholski, Laughlin, & Conway, 1999; Ericsson & Kintsch, 1995). Previous studies have shown that a substantial proportion of the variance related to solution accuracy in word problems is related to WM (see, e.g., LeBlanc & Weber-Russell, 1996; Swanson, in press; Swanson & Sachse-Lee, 2001). For example, LeBlanc and Weber-Russell (1996) found, via computer simulation and testing of children in Grade Levels K-3, that WM variables accounted for a substantial proportion of variance (between 49% and 57%) in children's word-problem solutions.

One purpose of this study was to identify cognitive processes and skills that underlie WM and word-problem-solving proficiency in three age groups (first-, second-, and third-grade students) in children at risk or not at risk for serious math difficulties (SMD). Our framework for isolating components of WM that are related to word-problem solving (as well as other domains) is Baddeley's (1986, 1996) multicomponent model. Baddeley (1986; Baddeley & Logie, 1999) described WM as a limited-capacity central executive system that interacts with a set of two passive store systems used for temporary storage of different classes of information: the speech-based phonological loop and the visual sketchpad. The phonological loop is responsible for the temporary storage of verbal information; items are held within a phonological store of limited duration and are maintained within the store via the process of articulation. The visual sketchpad is responsible for the storage of visual–spatial information over brief periods and plays a key role in the generation and manipulation of mental images. Both storage systems are in direct contact with the central executive system. The central executive system is considered to be primarily responsible for coordinating activity within the cognitive system but also devotes some of its resources to increasing the amount of information that can be held in the two subsystems (Baddeley & Logie, 1999).

DISTINCTIONS BETWEEN WORKING MEMORY AND SHORT-TERM MEMORY

The distinctions made between the central executive system and specific memory storage systems (i.e., the phonological loop) in some ways parallel the distinctions made between WM and short-term memory (STM). WM is referred to as a processing resource of limited capacity involved in the preservation of information

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while simultaneously processing the same or other information (see, e.g., Baddeley, 1986; Baddeley & Logie, 1999; Engle et al., 1999; Just & Carpenter, 1992). Individual differences in WM capacity have been attributed to executive processing (see, e.g., Engle et al. 1999; Swanson, 2003), such as the ability to inhibit irrelevant information (see, e.g., Chiappe, Hasher, & Siegel, 2000), as well as to speed of processing (Salthouse, 1996) and knowledge (see, e.g., Ericsson & Kintsch, 1995). In contrast, STM typically involves situations where small amounts of material are held passively (i.e., minimal resources from long-term memory [LTM] are activated to interpret the task, e.g., digit or word span tasks) and then reproduced in a sequential fashion. That is, participants are asked to only reproduce the sequence of items in the order they were presented (see, e.g., Daneman & Carpenter, 1980; Dempster, 1985; Klapp, Marshburn, & Lester, 1983).[1] Individual differences on these STM measures have been primarily attributed to phonological coding and rehearsal (see, e.g., Willis & Gathercole, 2001).

PHONOLOGICAL PROCESSING MODEL

How might WM mediate age-related and individual differences in word-problem solving? We tested two competing models as an explanation of the role of WM in age-related problem-solving performance in children at risk or not at risk for SMD. The first model hypothesizes that individual and age-related influence of WM on children's problem solving is primarily moderated by processing efficiency at the phonological level. A simple version of this hypothesis states that individuals at risk for SMD and younger children are slower and/or less accurate at processing verbal information (numbers, letters) than average-achieving children or older children and that such reduced processing on the participants' part underlies their poor WM and problem-solving performance. This hypothesis is consistent with a number of bottom-up models of reading (comprehension) that view the primary task of executive processing as one of relaying the results of lower level linguistic analyses upward through the language system (see, e.g., Shankweiler & Crain, 1986). Several studies have suggested that the phonological system, via the phonological loop (phonological store, subvocal rehearsal), influences verbatim memory capacity, which in turn supports comprehension (see, e.g., Perfetti, 1985). Likewise, some studies have attributed individual differences in mathematical problem solving to the phonological system (see Furst & Hitch, 2000; Swanson & Sachse-Lee, 2001, for reviews). This link occurs because mathematical word problems are a form of text, and the decoding and comprehension of text draw on the phonological system (see Shankweiler & Crain, 1986, for a review). In summary, the hypothesis assumes that phonologically analyzed information at word level or number level is transferred to WM storage, which in turn is transferred (thus freeing storage for the next chunk of phonological information) upward through the processing system to promote online extraction of meaning. Consistent with this assumption, extraction of meaning from text is compromised in children with SMD because inefficient phonological analysis creates a bottleneck that constricts information flow to higher levels of processing (see, e.g., Crain, Shankweiler, Macaruss, & Bar-Shalom, 1990).

What are potential measures of the phonological system? Several studies have assumed that STM measures capture a subset of WM performance, the utilization and/or operation of the phonological loop (see Gathercole, 1998; Gathercole & Baddeley, 1993, for comprehensive reviews). Some authors have suggested that the phonological loop may be referred to as verbal STM (see, e.g., Baddeley, 1986; Dempster, 1985) because it involves two major components discussed in the STM literature: a speech-based phonological input store and a rehearsal process (see Baddeley, 1986, for review).

Research to date suggests younger children rehearse less and perform more poorly on tasks requiring the short-term retention of order information than do older children (see, e.g., Ornstein, Naus, & Liberty, 1975), signifying inefficient utilization of the phonological rehearsal process (cf. Henry & Millar, 1993). Likewise, children with SMD have been found to suffer deficits in short-term retention when compared with children without SMD (see, e.g., Geary, Brown, & Samaranayake, 1991; Geary, Hoard, & Hamson, 1999; Siegel & Ryan, 1989).

There are clear expectations in the aforementioned model, specifically, that age-related and individual differences in children's problem solving are related to the phonological system. Thus, problem-solving proficiency, as well as math and reading performance, follows automatically with improvement in phonological processing. Specifically, if individual and age-related differences in WM and problem-solving performance are moderated by the phonological system, then the relationship between problem solving and WM should be eliminated when measures of the phonological system (e.g., STM, phonological awareness) are partialed from a statistical analysis.

EXECUTIVE PROCESSING MODEL

In contrast to the above model, the second model views executive processes as providing resources to lower order (phonological system) skills, as well as monitoring a general executive system independent of those skills (see, e.g., Baddeley & Logie, 1999). Given that the phonological loop is partly controlled by the central executive system, the development of problem solving may be directly related to the controlling functions of the central executive system itself. This model assumes there is variance that is unique to particular systems of WM (executive processing, phonological coding), as well as some shared variance with these systems (see Swanson & Alexander, 1997, for further discussion). Thus, in the context of WM development, the model suggests that

[1] Everyday examples of WM tasks would thus include holding a person's address in mind while listening to instructions about how to get there or perhaps listening to the sequence of events in a story while trying to understand what the story means. Everyday examples of STM tasks would include recalling a series of digits, such as a telephone number, in order immediately after their presentation. Although there is controversy concerning the nature of STM and WM tasks (see Engle et al., 1999, for review), there is some agreement that a transformation or active monitoring (e.g., focusing on relevant information when competing information is present) is required on WM tasks (see, e.g., Baddeley & Logie, 1999). For the sake of parsimony, in the present study, we view WM tasks as those that require some inference, transformation, and monitoring of relevant and irrelevant information, whereas STM tasks require the storage of information with minimal ongoing processing requirements that vary from initial encoding. Thus, tasks in the present study were selected according to the degree to which some monitoring of relevant and irrelevant information would be required prior to output.

both a general (executive) and specific (phonological) system contribute significant variance to individual and age-related differences in mathematical problem solving. In support of this model, Swanson and Sachse-Lee (2001) showed that when math knowledge and reading comprehension were controlled, phonological processing and WM each contributed unique variance to mathematical problem solving in math-disabled and non-math-disabled children in Grades 5 and 6. The results also showed that the entry of phonological processing first into a hierarchical regression model did not partial out the influence of WM on solution accuracy. These results suggest that fundamental processing problems in children's WM at the executive level play an important role in mediating accuracy in word-problem solving.

How might the executive system contribute to age-related and individual differences in problem solving? One of the possibilities we explored in this study relates to accurately accessing information from LTM (see, e.g., Cantor & Engle, 1993). Baddeley and Logie (1999) stated that a major role of WM "is retrieval of stored long-term knowledge relevant to the tasks at hand, the manipulation and recombination of material allowing the interpretation of novel stimuli, and the discovery of novel information or the solution to problems" (p. 31). They further stated, "any increase in total storage capacity beyond that of a given slave system is achieved by accessing either long-term memory (LTM) or other subsystems" (Baddeley & Logie, 1999, p. 37). Thus, the influence of WM performance on problem solving is related to one's ability to accurately access information (e.g., appropriate algorithm) from LTM to solve the problem. More specifically, a word problem introduces information into WM. The contents of WM are then compared with possible action sequences (e.g., associative links) in LTM (Ericsson & Kintsch, 1995). When a match is found (recognized), the contents of WM are updated and used to generate a solution. This assumption is consistent with current models of problem solving, which are based on recognize–act models of a cognitive processor (Anderson et al., 1996; Ericsson & Kintsch, 1995).

Although individual and age-related differences in problem-solving accuracy are possibly related to the retrievability of contents in LTM (e.g., knowledge of specific mathematical relations, general problem-solving strategies) accessed in WM, some research has suggested that other executive activities besides accessing information from LTM underlie the influence of WM solution accuracy. Also, several cognitive activities have been assigned to the central executive (e.g., see Miyake, Friedman, Emerson, Witzki, & Howerter, 2000, for a review), such as control of subsidiary memory systems, control of encoding and retrieval strategies, attention switching during manipulation of material held in the verbal and visual–spatial systems, suppression of irrelevant information, and so on, in addition to LTM knowledge retrieval (see, e.g., Baddeley, 1996; Miyake et al., 2000; Oberauer, Sub, Wilhelm, & Wittman, 2003). Recent studies have suggested that specific activities of the central executive related to suppression of irrelevant information and resource monitoring are deficient in children with math and/or reading disabilities (Bull & Scerif, 2001; Chiappe et al., 2000; Passolunghi, Cornoldi, & De Liberto, 1999; Swanson, 1999). A review of these studies (see Swanson & Siegel, 2001a, 2001b, for reviews) indicates that children with SMD and/or reading disabilities yield (a) poor performance on complex divided attention tasks; (b) poor monitoring, such as an inability to suppress (inhibit) irrelevant information; and (c) poor performance across verbal and visual–spatial tasks assumed to require both storage and processing when compared with normally achieving peers (also see Chiappe et al., 2000; De Beni, Palladino, Pazzaglia, & Cornoldi, 1998; Swanson, 1993).

PURPOSE AND PREDICTIONS

In summary, the purpose of this study was to assess the contribution of WM to individual and age-related performance in children's problem-solving performance. We considered two possible models: (a) that the relationship between WM and problem solving is primarily mediated by the phonological system or (b) that executive processes operate independent of the phonological system and therefore contribute unique variance to problem solving beyond the phonological system. Measures of the phonological system included tasks related to STM. Measures of the executive system were modeled after Daneman and Carpenter's (1980) WM tasks. These tasks demand the coordination of both processing and storage. Recent studies have suggested that these tasks capture at least two factors of executive processing: susceptibility to interference and manipulation of capacity (see, e.g., Oberauer, 2002; Whitney, Arnett, Driver, & Budd, 2001).

The major prediction in this study was that individual and age-related differences in WM would be partially mediated by an executive system that operates independent of the phonological system. Thus, individual and age-related changes in WM would be sustained when measures of phonological processing (e.g., STM, phonological knowledge) were partialed from the analysis. Variables of interest besides WM were computation knowledge, knowledge of processing operations, semantic processing, and reading. Each of these variables has been suggested as important to word-problem-solving accuracy (see Cooney & Swanson, 1990; Swanson, Cooney, & Brock, 1993; Swanson & Sachse-Lee, 2001, for reviews). We also investigated individual differences in children's recognition of the structural properties of word problems. Analogous to the structural properties outlined by Mayer and Hegarty (1996) and Cooney and Swanson (1990), four structural properties were investigated in the present study, specifically, the recognition of numerical, question, algorithmic knowledge, and irrelevant propositions.

In summary, three research questions directed this study.

1. Does WM predict problem solving after various measures of phonological processing have been partialed from the analysis?
2. Does the relationship between problem solving and WM vary as a function of age and risk for SMD?
3. What cognitive processes (e.g., LTM, inhibition) and skills (e.g., arithmetic, reading) mediate the relationship between WM and problem solving?

METHOD

Participants

Three hundred and fifty-three children from a southern California public school district and private school district participated in this study. Final

Table 1
Means and Standard Deviations for Measures as a Function of Children at Risk for Serious Math Difficulties (SMD) and Not at Risk (NSMD) as a Function of Age

	Grade 1 (*N* = 130)				Grade 2 (*N* = 92)				Grade 3 (*N* = 131)			
	SMD (*n* = 73)		NSMD (*n* = 57)		SMD (*n* = 34)		NSMD (*n* = 58)		SMD (*n* = 25)		NSMD (*n* = 106)	
Variable	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Classification												
Chronological age	6.19	0.49	6.23	0.46	7.41	0.50	7.29	0.50	8.52	0.51	8.20	0.49
Fluid intelligence—Raven												
Standard score	102.77	14.19	112.72	13.47	106.18	14.00	111.76	16.65	96.00	9.73	108.58	13.08
Raw score	18.37	5.12	22.25	5.46	23.50	4.94	25.83	5.83	23.64	4.52	27.70	4.95
Mental computation—WISC–III												
Standard score	7.04	2.85	13.10	2.20	7.90	1.77	11.78	2.18	5.87	2.18	11.72	2.08
Raw score	8.56	2.75	12.63	1.10	11.82	1.76	14.28	1.35	11.43	3.34	15.75	1.57
Digit-naming speed—CTOPP (standard score)	6.94	0.81	8.28	1.45	7.78	1.01	9.35	1.53	9.50	1.85	10.99	2.22
Criterion measures—Word problems												
Word-problem-solving processes (raw score)												
Question	1.46	0.91	2.12	0.80	1.68	0.87	1.97	0.86	1.80	1.04	2.42	0.80
Numbers	1.55	1.05	2.19	0.97	2.09	0.87	2.41	0.70	2.60	0.58	2.54	0.68
Goal	1.15	0.84	1.21	0.93	1.00	0.98	1.41	0.96	1.28	0.98	2.19	0.93
Operations	1.47	0.71	1.79	1.62	1.91	1.03	2.41	0.77	1.96	0.98	2.27	0.86
Algorithm	1.16	0.83	1.63	0.87	1.97	0.94	2.19	0.71	2.08	0.86	2.31	0.77
Irrelevant information	1.85	0.81	2.05	0.71	2.03	0.90	2.41	0.70	2.52	0.65	2.58	0.68
Total score	6.79	2.96	8.95	3.28	8.65	3.34	10.40	2.48	9.72	2.98	11.74	2.74
Word problems—Semantic structure varied (raw score)	8.62	3.23	8.62	3.23	10.71	3.45	12.81	2.65	12.24	3.26	14.29	3.06
Criterion measures—Math												
WRAT												
Standard score	110.97	11.39	115.97	10.15	99.00	9.36	106.41	9.79	107.04	14.57	117.68	12.12
Raw score	18.71	1.61	19.70	1.51	21.94	1.72	23.34	1.87	28.36	3.60	30.13	2.90
WIAT												
Standard score	102.63	13.18	110.91	13.59	100.79	11.28	109.36	10.64	108.28	13.46	116.25	9.97
Raw score	8.25	2.03	10.19	2.15	14.21	2.69	15.78	2.14	20.24	2.86	21.65	2.12
Computational fluency (raw score)	13.51	6.57	17.74	6.37	14.18	8.59	18.78	7.18	21.36	11.08	26.81	11.33
Prediction measures—Reading												
WRAT												
Standard score	102.89	16.70	116.61	14.77	98.24	12.62	108.67	12.78	95.17	16.83	107.11	11.54
Raw score	21.64	4.72	25.54	4.16	27.33	4.20	30.78	4.27	29.91	5.44	33.38	4.11
Reading—TOWRE real words (raw score)	21.64	13.80	37.56	14.68	44.33	13.18	58.17	10.35	54.74	16.29	65.67	8.94
Comprehensive—WRMT–R												
Standard score	100.78	13.19	111.46	10.34	102.12	9.83	108.03	11.12	96.26	13.06	104.96	8.77
Raw score	10.92	7.81	19.37	7.76	24.39	6.37	29.88	5.92	28.74	8.64	33.91	4.42
Rapid naming—Letters—CTOPP												
Standard (scale) score	6.83	0.84	8.16	1.28	7.81	0.94	9.17	1.45	9.39	1.88	10.17	1.80
Raw score	72.62	24.39	49.67	9.60	53.45	12.81	42.22	7.24	42.00	9.54	38.12	7.10
Phonological processing												
Pseudowords—TOWRE (raw score)	9.33	7.97	18.74	9.97	18.42	9.77	27.64	10.75	23.70	12.02	33.36	10.53
Elision (segmentation)—CTOPP (raw score)	5.26	3.98	10.16	4.82	9.64	5.24	12.38	4.87	10.04	6.01	13.47	5.40
Phonological fluency (raw score)	5.58	3.19	6.56	3.53	7.00	2.90	7.07	3.09	7.13	2.85	8.26	3.40
Short-term memory (highest span scores)												
Digits forward—WISC–III (span score)	2.44	1.00	2.82	0.97	3.06	0.75	3.19	0.89	2.52	0.85	3.33	1.06
Digits backward—WISC–III (span score)	1.00	0.60	1.37	0.62	1.30	0.59	1.49	0.57	1.30	0.63	1.66	0.77
Pseudoword span (span score)	3.38	0.91	4.32	2.49	4.58	1.85	5.16	2.84	3.87	2.10	4.93	2.20
Real-word span (span score)	3.33	0.91	3.75	0.74	3.67	0.99	4.09	0.68	3.70	0.70	3.98	0.72

Table 1 (*continued*)

Variable	Grade 1 (N = 130) SMD (n = 73) M	SD	NSMD (n = 57) M	SD	Grade 2 (N = 92) SMD (n = 34) M	SD	NSMD (n = 58) M	SD	Grade 3 (N = 131) SMD (n = 25) M	SD	NSMD (n = 106) M	SD
Working memory (highest span scores)												
Listening sentence span	1.26	0.57	1.60	0.80	1.76	0.78	2.05	0.78	1.61	0.83	2.14	0.75
Digit/sequence span—S-CPT	1.56	0.42	1.70	0.45	1.70	0.56	1.86	0.63	1.80	0.56	2.19	0.84
Semantic association—S-CPT	1.66	0.28	1.79	0.15	1.73	0.26	1.78	0.15	1.79	0.10	1.86	0.32
Visual matrix—S-CPT	2.52	1.19	2.72	1.16	2.82	1.27	3.28	1.23	3.40	1.29	3.44	1.29
Mapping/direction—S-CPT	1.44	0.35	1.39	0.31	1.30	0.28	1.41	0.33	1.43	0.43	1.52	0.57
Inhibition and updating												
Random-generation letters (computed score)	0.36	0.36	0.25	0.22	0.25	0.21	0.26	0.21	0.34	0.25	0.22	0.19
Random-generation numbers (computed score)	0.38	0.34	0.27	0.23	0.28	0.21	0.23	0.18	0.33	0.49	0.16	0.15
Updating (raw [proportion] score)	2.69	3.73	5.33	4.79	3.88	3.69	7.03	5.05	3.35	3.69	8.10	5.23
Semantic and vocabulary												
Semantic fluency (raw score)	9.25	3.76	12.93	4.56	12.88	3.98	13.71	3.88	13.70	4.49	15.25	4.19
Vocabulary												
Standard score	7.23	3.73	9.95	3.44	8.42	3.16	10.03	4.07	6.65	2.89	11.09	3.77
Raw score	10.29	4.98	13.93	4.56	16.18	4.79	18.29	6.39	17.13	4.71	23.87	6.63

Note. Raven = Raven Colored Progressive Matrices; WISC—III = Wechsler Intelligence Scale for Children—Third Edition; CTOPP = Comprehensive Test of Phonological Processing; WRAT = Wide Range Achievement Test; WIAT = Wechsler Individual Achievement Test; TOWRE = Test of Word Reading Efficiency; WRMT–R = Woodcock Reading Mastery Test—Revised; S-CPT = S-Cognitive Processing Test.

selection was related to parent approval for participation and achievement scores. Of the 353 children selected, 169 were girls, and 184 were boys. Gender representation was not significantly different among the three age groups, $\chi^2(2, N = 353) = 1.15$, $p > .05$. Ethnic representation of the sample was 163 Anglo, 147 Hispanic, 25 African American, 14 Asian, and 4 other (e.g., Native American, Vietnamese). The mean socioeconomic status (SES) of the sample was primarily middle class based on parent education or occupation. However, the sample varied from low-middle class to upper-middle class. Means and standard deviations for the selection variables used in this study are shown in Table 1.

Definition of Risk for Serious Math Difficulties

There are no generally agreed-upon criteria for defining children at risk for SMD, especially in Grade 1, where instruction is only beginning to address mathematical operations. Because first graders were used in our sample, we attempted to control in our classification tasks demands placed on reading and writing. In addition, our focus was on problem solving and not arithmetic calculation, and therefore, we used different measures (i.e., mental computation of word problems vs. paper-and-pencil computation of arithmetic problems) than studies that have defined math disabilities by computation skill. In contrast to the literature on math disabilities or reading disabilities, we assumed that children with SMD may not have skill difficulties related to arithmetic calculation or reading but may nevertheless have difficulties in coordinating arithmetic and language processes to solve a problem. Furthermore, because we were interested in the reasoning processes related to problem solving, we focused on a reliable measure that asked questions within a verbal context (e.g., If I have an apple and divide it in half, how many pieces do I have?) rather than a computational context ($1 + 1 = ?$). Thus, we utilized the oral presentation of story problems as a criterion measure of SMD.

In selecting children at risk, we also focused on a child's general fluency with numbers. We assumed that number-processing speed underlies children's ability to automatically access arithmetic facts, knowledge of meaning, signs, and procedures (Bull, Johnston, & Roy, 1999; Geary et al., 1991). We assumed that children who have quicker access to numbers, that is, faster number fluency, would be less at risk for mental computation difficulties than children less fluent in number naming. This seemed reasonable to us on the basis of Hitch and McAuley's (1991) finding that children with math difficulties evidenced deficits in the speed of implicit counting. Furthermore, speed of number naming has a parallel in the reading literature, where both letter-naming speed and phonological knowledge are assumed to underlie reading disabilities.

Thus, in this study, children at risk for SMD were defined as having normal intelligence (standard score > 85), but with performance below the 25th percentile (standard score of 90 or scaled score of 8) on standardized measures related to (a) solving orally presented word problems and (b) digit-naming fluency. The 25th-percentile cutoff score on standardized achievement measures has been commonly used to identify children at risk (see, e.g., Fletcher et al., 1989; Siegel & Ryan, 1989) and therefore was used in this study. Classification of children at risk (SMD) and not at risk (NSMD) was based on norm-referenced measures of computation on the arithmetic subtest of the Wechsler Intelligence Scale for Children—Third Edition (WISC–III; Wechsler, 1991) and digit-naming speed from the Comprehensive Test of Phonological Processing (CTOPP; Wagner, Torgesen, & Rashotte, 2000) described in the following section. Children who yielded scaled scores at or less than 8 on both measures were considered at risk for SMD. A scale score of 8 was equivalent to a standard score of 90 or a percentile score of 25.

In the present sample, 132 children were classified at risk for SMD. As expected, a larger proportion of children at risk for SMD were identified in Grade 1 (52%) than Grades 2 (35%) and 3 (22%), $\chi^2(2, N = 353) = 25.50$, $p < .0001$. No significant differences emerged between groups in terms of ethnicity, $\chi^2(5, N = 353) = 8.67$, $p > .05$, or gender, $\chi^2(1, N = 353) = .06$, $p > .05$. As expected, the portion of children who scored below the 25th percentile in word recognition was significantly higher for children at risk for SMD (25% of the SMD sample) than children not at risk for SMD (3%), $\chi^2(1, N = 353) = 44.47, p < .0001$.

Tasks and Materials

The battery of group- and individually administered tasks is described below. Experimental tasks are described in more detail than published and standardized tasks. Tasks were divided into classification, criterion, and predictor variables. Cronbach's alpha reliability coefficients for the sample were calculated for all measures and are provided.

Classification Measures

Fluid Intelligence

Fluid intelligence was assessed by the Raven Colored Progressive Matrices (Raven, 1976). Strong correlations (e.g., $r = .80$) have been noted between this measure and WM (see, e.g., Engle et al., 1999; Kyllonen & Christal, 1990). Children were given a booklet with patterns displayed on each page, each pattern revealing a missing piece. For each pattern, four possible replacement pattern pieces were displayed. Children were required to circle the replacement piece that best completed the patterns. After the introduction of the first matrix, children completed their booklets at their own pace. Patterns progressively increased in difficulty. The dependent measure (range of 0 to 36) was the number of problems solved correctly, which yielded a standardized score ($M = 100$, $SD = 15$). Cronbach's coefficient alpha was .88.

Mental Computation of Word Problems

This task was taken from the arithmetic subtest of the WISC–III (Wechsler, 1991). Each word problem was orally presented and was solved without paper or pencil. Questions ranged from simple addition (e.g., If I cut an apple in half, how many pieces will I have?) to more complex calculations (e.g., If three children buy tickets to the show for $6.00 each, how much change do they get back from $20.00?). The dependent measure was the number of problems solved correctly, which yielded a scaled score ($M = 10$, $SD = 2$). Cronbach's alpha for the WISC–III arithmetic subtest was .66 in the present study.

Digit-Naming Speed

The administration procedures followed those specified in the manual of the CTOPP (Wagner et al., 2000). For this task, the examiner presented participants with an array of 36 digits. Participants were required to name the digits as quickly as possible for each of two stimulus arrays containing 36 items, for a total of 72 items. The task administrator used a stopwatch to time participants on speed of naming. The dependent measure was the total time to name both arrays of numbers. The correlation between Array Forms A and B was .91.

Criterion Variables

Word-Problem Solving and Components

Mathematical word-problem-solving processes. This experimental test assessed the child's ability to retrieve processing components of word problems (Swanson & Sachse-Lee, 2001). Two booklets were adapted from Swanson et al. (1993) for students in Grades 2 and 3. A booklet was developed for students in Grade 1 that was conceptually consistent with the second- and third-grade experimental booklets but was grade-level appropriate in word and computational difficulty. Each booklet contained three problems that included pages assessing the recall of text from the mathematical word problems. The categories of mathematical word problems were addition, subtraction, and multiplication (the last for third graders only). Problems were four sentences in length and contained two assignment propositions, one relation, one question, and an extraneous proposition related to the solution. To control for reading problems, the examiner orally read (a) each problem and (b) all multiple-choice response options as the students followed along. For example, a subtraction problem read as follows: "Darren found 15 pinecones [assignment]. He threw 5 pinecones back [assignment]. Darren uses pinecones to make ornaments [extraneous]. How many pinecones did Darren keep [question]?" (The combination of Sentences 1 and 2 was the relation proposition.)

No titles were given to the problems except the titles Problem 1, Problem 2, and so on. Depending on the order of presentation, after the problem was read, students were then instructed to turn to the next page on which the following statement was written: "Without looking back at the problem, circle (from a choice of four options) the question the story problem was asking on the last page." The multiple-choice questions for the sample problem above were (a) How many pinecones did Darren have in all? (b) How many pinecones did Darren start with? (c) How many pinecones did Darren keep? and (d) How many pinecones did Darren throw back? This page assessed student ability to correctly identify the question proposition of each story problem.

On the next page for each problem, directions asking, "Without looking back at the problem, try to identify the numbers in the problem" were read. The multiple-choice questions for the sample problem above were (a) 15 and 5, (b) 5 and 10, (c) 15 and 20, and (d) 5 and 20. This page assessed student ability to correctly identify the numbers in the two assignment propositions of each story problem.

Instructions on the next page were read as follows: "Without looking back at the problem, identify what the question wants you to find." The multiple choice questions were (a) the total number of pinecones Darren found all together, (b) what Darren plans to do with the pinecones, (c) the total number of pinecones Darren had thrown away, and (d) the difference between the pinecones Darren kept and the ones he threw back. This page assessed the student's ability to correctly identify the goals in the two assignment propositions of each story problem.

Instructions for the final page were "Without looking back at the problem, identify whether addition, subtraction, or multiplication was needed to solve the problem." Students were directed to choose one of the two or three operations: (a) addition (b) subtraction, and (c) multiplication (for third graders only). After choosing one of the two or three operations, children were then asked to identify the number sentence they would use to solve the problem: (a) $15 \times 5 =$, (b) $15 + 10 =$, (c) $15 - 5 =$, or (d) $15 + 5 =$. This page of the booklet assessed the student's ability to correctly identify the operation and algorithm, respectively.

At the end of each booklet, students were read a series of true–false questions. All statements were related to the extraneous propositions for each story problem within the booklet. For example, the statement "Darren used pinecones to make ornaments" would be true, whereas the statement "Darren used pinecones to draw pictures" would be false. The total score possible for propositions related to question, number, goal, operations, algorithms, and true–false questions was 12. Cronbach's alpha for the experimental word-problem-solving booklet task was .77.

Word problems—Semantic structure varied. The purpose of this experimental measure was to assess mental problem solving as a function of variations in the semantic structure of a word problem. Children were orally presented the problem and asked to calculate the answer in their head. The word problems were derived from the work of Riley, Greeno, and Heller (1983); Kintsch and Greeno (1985); and Fayol, Abdi, and Gombert (1987). There were four sets of questions. Eight questions within each set were ordered by the difficulty of responses.

The first set of eight questions focused on *change problems*. The solution difficulty of problems in this set involved sums of 9 or less. An example change problem is "Paul had 5 candies. His mother gave him 2 candies. How many candies does Paul now have?"

Although there were systematic variations in the presentation of each change problem within the set, the computation difficulty of the problem stayed the same. For example, the first question followed a standard format in which a result was unknown. For example, "Alan had 5 marbles. His friend gave him 2 marbles. His sister then gave him 2 marbles. How many marbles does Alan have?"

The second question, when compared with the first problem above, changed the location of the question. The question was presented in the first sentence. For example, "How many marbles does Alan have? Alan had 5 marbles. His friend gave him 2 marbles. His sister then gave him 2 marbles."

The third question introduced a sequence of events. For example, "This morning Alan had 5 marbles in his pocket. At noon his friend gave him 2 marbles. Yesterday his sister gave him 2 marbles. How many marbles does Alan have?"

The fourth question changed both the location of the question and the temporal sequence of the question. For example, "How many marbles does Alan have? This morning Alan had 5 marbles in his pocket. At noon his friend gave him 2 marbles. Later in the day his sister gave him 2 marbles."

The next four questions followed the same format as the first four questions (standard, location of question, time sequence, location of question and time sequence), except the word *now* was introduced in either the first or second sentence. This change in semantic structure differed from the first four questions because the child had to determine the initial state of a problem. For example, "Alan has now 6 marbles. During morning playtime at school, he won 2 marbles. At noon he won 2 marbles. How many marbles did Alan have before he went to school this morning?" The introduction of the word *now* told the child the result. However, the child had to determine the start of the question (initial state). The *now* in the first sentence provided the child with the final state of the problem.

The second set of eight questions followed the same format of questions in Set 1 except that the solution difficulties involved sums greater than 10 but less than 20.

The third set of questions focused on *compare questions* rather than change questions. Compare questions are more difficult than change questions (Riley et al., 1983). Compare questions focus on differences in quantity, for instance, how many more or how many less. For example, "Paul has 5 candies. Paul has 2 less candies than his sister. How many candies does his sister have?" Word problems followed the same format as the other sets (standard, location of question, time sequence, location of question and time sequence, initial state) and minuends of 9 or less. Set 4 included the same types of problems as Set 3, but problems had minuends between 10 and 20.

Because the questions varied in difficulty, not all questions were administered. Questions in Set 1 were administered first. All eight questions were administered unless three errors occurred. If two or fewer errors were made, participants were administered Set 2 and so on. The dependent measure was the number of problems solved correctly. The total possible number of correct solutions was 32. Cronbach's alpha for the word-problem-comprehension task was .82.

Arithmetic Calculation

Arithmetic computation. The arithmetic subtests from the Wide Range Achievement Test (WRAT; Wilkinson, 1993) and the Wechsler Individual Achievement Test (WIAT; Psychological Corporation, 1992) were administered. Both subtests required written computation for problems that increased in difficulty. Problems began with simple calculations ($2 + 2 =$) and moved up to algebra. The dependent measure was the number of problems solved correctly, which yielded a standard score ($M = 100, SD = 15$). Cronbach's alpha for the WRAT was .92 and for the WIAT was .93.

Computation fluency. This test was adapted from the Test of Computational Fluency (Fuchs, Fuchs, Eaton, Hamlett, & Karns, 2000). The adaptations required students to write answers within 2 min to 25 basic facts and algorithms for Grades 1, 2, and 3. The basic facts and algorithms were problems matched to grade level. The dependent measure was the number of problems solved correctly. Cronbach's alpha was .85.

Predictor Variables

Reading and Phonological Processing Measures

Because the phonological measures are commonly used and derived from published standardized measures (i.e., Woodcock Reading Mastery Test—Revised [WRMT–R; Woodcock, 1998], WISC–III, CTOPP [Wagner et al., 2000]), we only briefly describe these tasks.

Real-word and pseudoword reading tasks. Two subtests were administered from the Test of Word Reading Efficiency (TOWRE; Wagner & Torgesen, 1999). The two subtests required oral reading of a list of 120 real words or pseudowords of increasing difficulty. Real-word reading efficiency was assessed by the Sight Word Efficiency subtest. Students were given 45 s to read aloud as many words as possible from a list of common words. Pseudoword reading ability was assessed by the Nonword Efficiency subtest. Students were given 45 s to read aloud as many words as possible from a list of nonwords. The nonwords followed regular spelling patterns, requiring students to quickly decipher pronunciations on the basis of their existing knowledge of grammar. The dependent measures for both subtests were the number of words read correctly in 45 s. Cronbach's coefficient alpha for the Sight Word Efficiency subtest was .90 and for the pseudoword subtest was .88.

Word recognition. Word recognition was assessed by the reading subtest of the WRAT. The task provided a list of words of increasing difficulty. The child's task was to read the words until 10 errors occurred. The dependent measure was the number of words read correctly. Cronbach's alpha for the word-recognition task was .89.

Reading comprehension. Reading comprehension was assessed by the Passage Comprehension subtest from the WRMT–R (Woodcock, 1998). The purpose of this task was to assess the child's comprehension of topic or subject meaning during reading activities. Comprehension questions were drawn from the reading of short paragraphs. The dependent measure was the number of questions answered correctly. Cronbach's coefficient alpha was .90.

Letter-naming speed. The administration procedures followed those specified in CTOPP (Wagner et al., 2000), including the presentation of practice trials. The manual reported correlations between parallel forms ranging from .80 to .93. For this task, the examiner presented participants with an array of 36 letters. Participants were required to name the letters as quickly as possible for each of two stimulus arrays containing 36 letters, for a total of 72 letters. The task administrator used a stopwatch to time participants on speed of naming. The dependent measure was the total time to name both arrays of letters. The correlation between Array Forms A and B was .90.

Phonological deletion. The Elision subtest from the CTOPP (Wagner et al., 2000) was administered. The Elision subtest measures the ability to parse and synthesize phonemes. The child was asked to say a word and to say what word is left if part of the word is deleted. For example, "Say heat. Now say the word if I said heat without saying the /t/." There were four practice items and 15 test items. The dependent measure was the number of items said correctly. Cronbach's coefficient alpha was .94.

Phonological fluency. This experimental measure was adapted from Harrison, Buxton, Husain, and Wise (2000). Children were given 60 s to generate as many words as possible beginning with the letter B. Children were told, "I want to see how many words you can say that begin with a certain letter. Do not say proper nouns or numbers or the same word with different endings, and try not to repeat yourself. Keep naming words that start with the letter until I say, 'Stop.' Speak clearly and loud enough so that I can hear the word you are saying. Do you understand? The letter is B, begin." Repetitions, proper name errors, and contravention of the stem repetition were deleted from the analysis. The dependent measure was the number of words correctly stated in 60 s. Cronbach's coefficient alpha was .88.

Short-Term Memory Measures

Four measures of STM were administered: forward digit span and backward digit span, word span, and pseudoword span. The digit subtest from the WISC–III was administered. The forward and backward parts of this subtest were maintained as separate variables rather than combining them to create a composite STM span. The forward digit span task required participants to recall and repeat in order sets of digits that had been spoken by the examiner and that increased in number. The technical manual reported a test–retest reliability of .91. The backward digit span task required participants to recall in reverse order sets of digits administered in the same manner as the forward digit span task. The reliability reported for this task was .76. Dependent measures for both tasks were the highest set of items recalled in order (range of 0 to 8 for digits forward; range of 0 to 7 for digits backward). It was assumed that forward spans presumably involved a subsidiary memory system (the phonological loop). However, the backward span task was assumed to involve a subsidiary system (phonological), as well as some resources from the executive system. It was reasoned that if a backward span test represented additive effects of a subsidiary system plus attention demands or control processes representative of a central executive system, then it should load with the WM tasks in the factor analysis (to be discussed below). However, if the backward digit span task primarily reflected a subsidiary system, it would load in the factor analysis with the STM measures. Cronbach's alpha for both the forward and backward tasks was .84.

The word span and pseudoword span tasks were presented in the same manner as the forward digit span measure. The word span task was previously used by Swanson, Ashbaker, and Lee (1996). The word stimuli were one- or two-syllable high-frequency words. Students were read lists of common but unrelated nouns and then were asked to recall the words. Word lists gradually increased in set size from a minimum of two words to a maximum of eight. The phonetic memory task (pseudoword span task; Swanson & Berninger, 1995) used strings of nonsense words (one syllable long), which were presented one at a time in sets of two to six nonwords (e.g., *DES, SEEG, SEG, GEEZ, DEEZ, DEZ*). The dependent measure for all STM measures was the highest set of items retrieved in the correct serial order (range of 0 to 7). Cronbach's alpha was .62 for the word span task and .82 for the phonetic memory task.

Working Memory Measures

The WM tasks in this study required children to hold increasingly complex information in memory while responding to a question about the task. The questions served as distractors to item recall because they reflected the recognition of targeted and closely related nontargeted items. A question was asked for each set of items, and the tasks were discontinued if the question was answered incorrectly or if all items within a set could not be remembered. Thus, WM span reflected a balance between item storage and correct responses to questions. Consistent with a number of previous studies, our WM tasks required the maintenance of some information during the processing of other information. For example, consistent with Daneman and Carpenter's (1980) seminal WM measure, the processing of information was assessed by asking participants simple questions about the to-be-remembered material (storage plus processing demands), whereas storage was assessed by accuracy of item retrieval (storage demands only). The question required a simple recognition of new and old information and was analogous to the yes–no response feature of Daneman and Carpenter's task. It is important to note, however, that in our tasks, the difficulty of the processing question remained constant within task conditions, thereby allowing the source of individual differences to reflect increased storage demands. Furthermore, the questions focused on the discrimination of items (old and new information) rather than deeper levels of processing such as mathematical computations (see, e.g., Towse, Hitch, & Hutton, 1998). A previous study with a different sample had established the reliability and the construct validity of the measures with the Daneman and Carpenter measure (Swanson, 1996). For this study, four WM tasks were divided into those requiring the recall of verbal (sentence/digit task, semantic association task) and visual–spatial information (e.g., visual matrix task, mapping/direction task) and were selected from a standardized battery of 11 WM tasks because of their high construct validity and reliability (see Swanson, 1992). The complete description of administration and scoring of the tasks is reported in Swanson (1995). A children's adaptation of the Daneman and Carpenter measure (Swanson, 1992) was also administered. Task descriptions follow.

Listening sentence span. The children's adaptation (Swanson, 1992) of Daneman and Carpenter's (1980) sentence span task was administered. The construction of and pattern of results associated with the two measures are comparable. The only difference was that each sentence was read to the child with a 5-s pause that indicated the end of a sentence. The original sentence span measure was used with university students, whereas the current measure used a simpler sentence structure and reading vocabulary. As a common measure of WM (see Daneman & Carpenter, 1980; Just & Carpenter, 1992), this task required the presentation of groups of sentences, read aloud, for which children simultaneously tried to understand the passage and remember the last word of each sentence. The number of sentences in the group gradually increased. After each group, the participant answered a question about a sentence and then recalled the last word of the sentence. WM capacity was defined as the largest group of ending words recalled. The mean sentence-reading level was approximately 3.8. The dependent measure was the highest set recalled correctly (range of 0 to 8) in which the process question was answered correctly. Cronbach's coefficient alpha was .79.

Semantic association task. The purpose of this task was to assess the participant's ability to organize sequences of words into abstract categories (Swanson, 1992, 1995). The participant was presented a set of words (one every 2 s), asked a discrimination question, and then asked to recall the words that "go together." For example, a set might include the following words: *shirt, saw, pants, hammer, shoes, nails*. Participants were directed to retrieve the words that went together (i.e., *shirt, pants,* and *shoes*; *saw, hammer,* and *nails*). The discrimination question was "Which word, *saw* or *level,* was said in the list of words?" Thus, the task required participants to transform information encoded serially into categories during the retrieval phase. The range of set difficulty was from two categories of two words to five categories of four words. The dependent measure was the highest set recalled correctly (range of 0 to 8) in which the process question was answered correctly. Cronbach's coefficient alpha was .85.

Digit/sentence span. This task assessed the child's ability to remember numerical information embedded in a short sentence (Swanson, 1992, 1995). Before stimulus presentation, the child was shown a card depicting four strategies for encoding numerical information to be recalled. The pictures portrayed the strategies of rehearsal, chunking, association, and elaboration. The experimenter described each strategy to the child before administration of targeted items. After all strategies had been explained, the child was presented numbers in a sentence context. For example, Item 3 stated, "Now suppose somebody wanted to have you take them to the supermarket at 8 6 5 1 Elm Street?" The numbers were presented at 2-s intervals, followed by a process question, for instance, "What was the name of the street?" Then, the child was asked to select a strategy from an array of four strategies that represented the best approximation of how he or she planned to practice the information for recall. Finally, the examiner prompted the child to recall the numbers from the sentence in order. No further information about the strategies was provided. Students were allowed 30 s to remember the information. Recall difficulty for this task ranged from 3 digits to 14 digits; the dependent measure was the highest set correctly recalled (range of 0 to 9) in which the process question was answered correctly. Cronbach's coefficient alpha was .79.

Visual matrix task. The purpose of this task was to assess the ability of participants to remember visual sequences within a matrix (Swanson, 1992, 1995). In contrast to the standardization procedures (Swanson, 1995), the visual matrix task was administered in small groups. An overhead projector was used to display stimuli to groups of children instead of individually by use of the examiner's manual. This change in format required students to circle their answer to the process question, rather than verbally responding. Otherwise, the task was administered as per the manual instructions. Participants were presented a series of dots in a matrix and were allowed 5 s to study the matrix. The matrix was then removed, and participants were asked, "Are there any dots in the first column?" To ensure the understanding of columns prior to test, participants were shown the first column's location and practiced finding it on blank matrices. In addition, for each test item, the experimenter pointed to the first column on a blank matrix (a grid with no dots) as a reminder of the first column's location. After answering the discriminating question (by circling *y* for yes or *n* for no), students were asked to draw the dots they remembered seeing in the corresponding boxes of their blank matrix response booklets. The task difficulty ranged from a matrix of 4 squares and 2 dots to a matrix of 45 squares and 12 dots. The dependent measure was the highest set recalled correctly (range of 0 to 11) in which the process question was answered correctly. Cronbach's alpha was .42.

Mapping and directions. This task required the child to remember a sequence of directions on a map (Swanson, 1992, 1995). The experimenter presented a street map with dots connected by lines; arrows illustrated the direction a bicycle would go to follow this route through the city. The dots represented stoplights, and lines and arrows mapped the route through the city. The child was allowed 10 s to study the map. After the map was removed, the child was asked a process question, for example, "Were there any stoplights on the first street (column)?" The child was then presented a blank matrix on which to draw the street directions (lines and arrows) and stoplights (dots). Difficulty ranged on this subtest from 4 dots to 19 dots. The dependent measure was the highest set of a correctly drawn map (range of 0 to 9) for which the process question was answered correctly. Cronbach's alpha was .94.

Inhibition and Updating Measures

Random generation of letters and numbers. The random-generation task has been well articulated in the literature (e.g., Baddeley, 1996; Towse, 1998). The task was assumed to measure inhibition because participants were required to actively monitor candidate responses and suppress responses that would lead to well-learned sequences, such as 1–2–3–4 or a–b–c–d (Baddeley, 1996). Because this task has been primarily used with adult samples who have quicker access to letters and numbers, it was modified for the age groups in this study. Each child was asked to write as quickly as possible numbers (or letters) first in sequential order to establish a baseline. Children were then asked to quickly write numbers (or letters) in a random, nonsystematic order. For example, for the number section, students were first asked to write numbers from 0 to 9 in order (i.e., 1, 2, 3, 4, 5, 6, 7, 8, 9) as quickly as possible in a 30-s period. They were then asked to write numbers as quickly as possible out of order in a 30-s period. Scoring included an index for randomness, information redundancy, and percentage of paired responses to assess the tendency of participants to suppress response repetitions. The measure of inhibition was calculated as the number of sequential letters or numbers minus the number of correctly unordered numbers or letters divided by the number of sequential letters or numbers plus the number of unordered letters or numbers. Cronbach's alpha for the random number-generation task in the current sample was .89 and for random letter generation was .91.

Updating. The experimental updating task was adapted from Morris and Jones (1990). A series of one-digit numbers was presented that varied in set lengths of 9, 7, 5, and 3. No digit appeared twice in the same set. The examiner told the child that the length of each list of numbers might be 3, 5, 7, or 9 digits. Participants were then told that they should recall only the last three numbers presented. Each digit was presented at approximately 1-s intervals. After the last digit was presented, the participant was asked to name the last three digits in order. It was stressed that some of the lists of digits would be only three digits long so the participants should not ignore any items. In contrast to the aforementioned WM measures, which involved a dual-task situation where participants answered questions about the task while retaining information (words or spatial location of dots), the current task involved the active manipulation of information such that the order of new information was added to or replaced the order of old information. That is, to recall the last three digits in an unknown ($N = 3$, 5, 7, 9) series of digits, the order of old information (previously presented digits) had to be kept available along with the order of newly presented digits. Thus, task performance reflected the activity of both the phonological system and the executive system. The dependent measure was the total number of digits correctly repeated (range of 0 to 16). Cronbach's alpha for the current sample was .94.

Semantic Processing and Vocabulary

Semantic fluency. The experimental measure was adapted from Harrison et al. (2000). Children were given 60 s to generate as many names of animals as possible. Children were told, "I want to see how many animals you can name. Try not to repeat yourself. Don't say pet names. Keep going until I tell you to stop. Ready, begin." Repetitions were deleted from the analysis. The dependent measure was the number of words correctly stated within 60 s. The coefficient alpha for the experimental semantic fluency task was .91.

Vocabulary. A word-knowledge score was obtained from the WISC–III vocabulary subtest. Children were read a word by the examiner and were asked to provide the meaning of the word. Words increased in complexity. Scoring followed the WISC–III manual. The dependent measure was based on the quality of word definitions. The coefficient alpha for the current sample was .88.

Procedures

Three doctoral-level graduate students trained in test administration tested all participants in their schools. Two sessions of approximately 45 to 60 min each were required for small-group test administration and one session of 45 to 60 min for individual administration. During the group-testing session, data were obtained from the Raven Colored Progressive Matrices, WIAT, WRAT, mathematical word-problem-solving processes task (WPS-P), visual matrix test, and arithmetic calculation fluency. The remaining tasks were administered individually. Test administration was counterbalanced to control for order effects. Task order was random across participants within each test administrator.

RESULTS

The means and standard deviations for intelligence, accuracy in recognizing problem-solving components, problem-solving solution accuracy, arithmetic calculation, phonological processing, reading, STM, WM, inhibition/updating, and semantic processing and vocabulary are shown in Table 1. The analyses and results were divided into two sections. The first section focused on age and ability group differences. SMD children and NSMD children were compared across three grade levels (first, second, and third grades). This approach lent itself to comparing age and ability group but had the disadvantage of ignoring information about the variability of participants in each group. The second section focused on correlations between WM and problem solving in the complete sample. The approach allowed us to study the entire range of scores in WM and problem solving as well to focus on common and unique variance. Regression models were computed to isolate unique processes that underlie word-problem-solving performance.

Age and Ability Group Comparisons

Preliminary Analysis

Prior to the analysis, we compared the age and risk groups on measures related to fluid intelligence (IQ). A 3 (age) × 2 (group risk: SMD vs. NSMD) analysis of variance (ANOVA) was computed on standard scores from the Raven Colored Progressive Matrices. A shown in Table 2, a significant main effect emerged for ability group. Thus, standard scores from the Raven Colored Progressive Matrices test were used as a covariate in the subsequent analysis. ANOVAs were also computed on the classification measures related to the arithmetic subtest of the WISC–III and the rapid number naming subtest from the CTOPP. As shown in Table 2, high effect sizes emerged between the two ability groups on these measures. Table 2 shows the univariates and effect sizes (η^2) for each comparison. An η^2 of .13, .05, or .02 corresponded to a Cohen's *d* of .80, .50, or .20, respectively.

A series of 3 (age) × 2 (group risk: SMD vs. NSMD) factorial multivariate analyses of covariance (MANCOVAs) was conducted to examine differences in means for the following processes: (a) word-problem components, (b) arithmetic calculation, (c) reading, (d) phonological processing, (e) STM, (f) WM, (g) inhibition/updating, and (h) semantic processing and vocabulary. Results related to the univariate analyses of covariance (ANCOVAs) are shown in Table 2. Because of the number of comparisons, a significant alpha level of $p < .001$ was adopted to control for

Table 2
Analysis of Covariance (ANCOVA) Word-Problem Solving, Calculation, Reading, and Cognitive Measures

Variable	Age *F* ratio	Age η^2	Group *F* ratio	Group η^2
Classification				
Chronological age	571.53†	.61	3.29	.009
Fluid Intelligence–Raven				
Standard score	1.90	.005	31.97†	.08
Raw score	60.18†	.15	30.88†	.08
Arithmetic subtest—WISC–III				
Standard score	6.16	.017	399.43†	.53
Raw score	185.4†	.34	251.1†	.41
Digit-naming speed—CTOPP				
Standard score	124.77†	.26	56.06†	.14
Raw score	134.43†	.27	95.63†	.21
Criterion—Word-problem solving				
Word-problem solving processes (raw score)				
Question	6.20	.01	13.98†	.04
Numbers	20.09†	.05	4.59	.01
Goal	14.05†	.03	7.95	.02
Operations	11.20†	.03	3.39	.01
Algorithm	35.10†	.09	2.95	.01
Irrelevant information	17.60†	.05	3.22	.01
Word problems—Semantic structure varied	43.64†	.11	15.01†	.04
Math calculation				
WRAT	519.30†	.59	13.79†	.04
WIAT	785.90†	.69	19.90†	.05
Computational fluency	23.03†	.06	13.12†	.04
Predictor: Reading				
Word recognition—WRAT	115.42†	.24	28.62†	.08
Real-word fluency—TOWRE	200.42†	.36	54.65†	.13
Predictor: Reading (*continued*)				
Comprehension—WRMT–R	222.25†	.38	35.91†	.09
Letter rapid naming—CTOPP	62.45†	.14	50.13†	.12
Phonological				
Pseudowords—TOWRE	64.11†	.15	37.93†	.09
Elision—CTOPP	24.13†	.06	21.68†	.06
Phonological fluency	7.01***	.02	1.79	.005
Short-term memory				
Digits forward—WISC–III	6.95	.02	15.04***	.04
Digits backward—WISC–III	5.97	.02	11.37†	.03
Pseudoword span	4.99	.01	5.61	.02
Real-Word span	6.33	.02	9.77***	.03
Working memory				
Listening sentence span	15.81†	.04	8.90	.03
Digit/sequence span—S-CPT	9.37†	.02	4.94	.02
Semantic association—S-CPT	3.99	.01	4.78	.02
Visual matrix—S-CPT	11.20†	.03	.77	.002
Mapping/directions—S-CPT	2.40	.006	0.11	.001
Inhibition and updating				
Random-generation letters	.75	.002	4.32	.01
Random-generation numbers	8.23†	.02	5.31	.02
Updating	5.42	.02	25.44†	.08
Semantic and vocabulary				
Semantic fluency	18.71†	.05	11.85†	.03
Vocabulary	70.81†	.16	18.21†	.05

Note. ANCOVAs were computed for raw scores on all but the four Classification tasks. Effect sizes are based on *F* ratios with influence of Raven scores partialed out. Raven = Raven Colored Progressive Matrices; WISC–III = Wechsler Intelligence Scale for Children—Third Edition; CTOPP = Comprehensive Test of Phonological Processing; WRAT = Wide Range Achievement Test; WIAT = Wechsler Individual Achievement Test; TOWRE = Test of Word Reading Efficiency; WRMT–R = Woodcock Reading Mastery Test—Revised; S-CPT = S-Cognitive Processing Test.
*** $p < .001$. † $p < .0001$.

experimentwise alpha inflation and Type I error. None of the Age × Group interactions for the MANCOVAs met the .001 alpha level, and therefore, the F ratios related to the interactions are not reported.

Word-Problem-Solving Components and Accuracy

A 3 (age) × 2 (group risk: SMD vs. NSMD) MANCOVA was conducted examining the six components of word-problem solving, identifying (a) the question, (b) the numerical information, (c) the goal of the problem, (d) the arithmetical operation, (e) the algorithm, and (f) irrelevant information. These measures were taken from the WPS-P. A significant multivariate main effect emerged for age, Wilks's $\Lambda = .72$, $F(12, 682) = 10.07$, $p < .0001$, $\eta^2 = .28$, and for group, Wilks's $\Lambda = .94$, $F(6, 341) = 6.56$, $p < .0001$, $\eta^2 = .06$. The overall pattern for the MANCOVA, which tests for the linear combination of the variables (determined by variable intercorrelations), is that students in third grade scored higher on all WPS-P components than those in second and first grades and that children not at risk scored higher than those at risk for SMD. The results of the ANCOVAs showed that five WPS-P components (identifying number, goal, operations, algorithms, and irrelevant information) differed by age. The largest effect size as a function of age was knowledge of algorithms. Only one ANCOVA (identify question) was significantly related to ability group. Overall, effect sizes related to age and ability group were in the low to moderate range.

An ANCOVA was computed on the solution-accuracy scores for the word-problem-solving measure that varied the semantic structure of the sentences. As shown in Table 2, the ANCOVA was significant for age and ability group. The results showed that students in Grade 3 scored higher than those in the earlier grades and that children not at risk scored higher on measures related to word-problem accuracy than those at risk for SMD.

Arithmetic

A 3 (age) × 2 (group risk: SMD vs. NSMD) factorial MANCOVA was conducted on accuracy scores for arithmetic measures (WRAT—math, WIAT—math, computation fluency). A significant multivariate main effect emerged for age, Wilks's $\Lambda = .13$, $F(6, 688) = 194.44$, $p < .0001$, $\eta^2 = .87$, and for group, Wilks's $\Lambda = .89$, $F(3, 344) = 12.88$, $p < .0001$, $\eta^2 = .11$. As expected, the general pattern of results is that students in Grade 3 scored higher than those in Grades 1 and 2 and that children not at risk scored higher on arithmetic measures than those at risk for SMD. It is important to note, however, that although a clear advantage was found for children not at risk, standard scores for the children at risk were in the normal range, suggesting that mathematical problem solving (as reflected by low standard scores on the arithmetic subtest of the WISC–III) shares some independence from calculation skill.

Reading Measures

A 3 (age) × 2 (group risk: SMD vs. NSMD) MANCOVA was conducted on accuracy scores for reading (WRAT—reading, WRMT–R—reading comprehension, TOWRE—real words, letter-naming speed). A significant multivariate effect emerged for age, Wilks's $\Lambda = .40$, $F(8, 678) = 49.56$, $p < .0001$, $\eta^2 = .60$, and for group, Wilks's $\Lambda = .82$, $F(4, 339) = 19.92$, $p < .0001$, $\eta^2 = .18$. The general pattern of the results is that students in Grade 3 scored higher than those in Grades 1 and 2 and that children not at risk scored higher on reading and phonological processes measures than those at risk for SMD.

Phonological Measures

A 3 (age) × 2 (group risk: SMD vs. NSMD) MANCOVA was conducted on accuracy scores for phonological measures (TOWRE—pseudowords, elision, phonological fluency). A significant multivariate effect emerged for age, Wilks's $\Lambda = .71$, $F(6, 682) = 21.00$, $p < .0001$, $\eta^2 = .29$, and for group, Wilks's $\Lambda = .89$, $F(3, 341) = 14.92$, $p < .0001$, $\eta^2 = .11$. The general pattern of the results is that students in Grade 3 scored higher than those in Grades 1 and 2 and that children not at risk scored higher on reading and phonological processes measures than those at risk for SMD.

Short-Term Memory

A 3 (age) × 2 (group risk: SMD vs. NSMD) MANCOVA was conducted on measures related to memory using span scores for STM (forward digit, backward digit, pseudoword span, real-word span). A significant multivariate main effect emerged for age, Wilks's $\Lambda = .90$, $F(8, 678) = 4.25$, $p < .0001$, $\eta^2 = .10$, and for group, Wilks's $\Lambda = .92$, $F(4, 339) = 7.70$, $p < .0001$, $\eta^2 = .08$. Students in third grade scored higher on memory measures than those in second and first grades, and children not at risk scored higher than those at risk for SMD. The ANCOVAs show that significant age effects emerged for only the digit forward task. In contrast, significant ANCOVAs as a function of ability group emerged on the digit forward, digit backward, and real-word span tasks. Effect sizes for the significant ability group ANCOVAs were in the low to moderate range.

Working Memory

A 3 (age) × 2 (group risk: SMD vs. NSMD) MANCOVA was conducted on measures related to span scores for WM (listening sentence span, digit/sentence, semantic association, visual matrix, mapping/directions). A significant multivariate main effect emerged for age, Wilks's $\Lambda = .82$, $F(10, 687) = 6.62$, $p < .0001$, $\eta^2 = .18$, and for group, Wilks's $\Lambda = .95$, $F(5, 339) = 3.10$, $p < .001$, $\eta^2 = .05$. Students in third grade scored higher on memory measures than those in second and first grades, and children not at risk scored higher than those at risk for SMD. The ANCOVAs showed age effects on all measures except for the semantic association and the mapping/directions tasks (all effect sizes were in the moderate range). ANCOVAs for isolated tasks as a function of ability group were nonsignificant at the .001 level. Furthermore, effect sizes as a function of ability group were in the low to moderate range.

Random Generation and Updating

A 3 (age) × 2 (group risk: SMD vs. NSMD) MANCOVA was conducted on measures of random generation and updating (random generation of letters and random generation of numbers, updating task). Rapid naming of numbers was not included because it served as a classification measure. A significant multivariate main effect emerged for age, Wilks's $\Lambda = .92$, $F(6, 678) = 4.35$, $p < .0001$, $\eta^2 = .08$, and for group, Wilks's $\Lambda = .91$, $F(3, 339) = 10.64$, $p < .0001$, $\eta^2 = .09$. As shown in Table 1, students in third grade scored higher than those in second and first grades, and students not at risk for SMD scored higher than students at risk for SMD. Significant ANCOVAs as a function of age emerged on the random generation of numbers task and as a function of ability group on the updating task.

Semantic Fluency and Vocabulary

A 3 (age) × 2 (group risk: SMD vs. NSMD) MANCOVA was conducted on vocabulary and semantic fluency measures (WISC–III Vocabulary subtest and semantic fluency). A significant multivariate main effect emerged for age, Wilks's $\Lambda = .69$, $F(4, 684) = 34.23$, $p < .0001$, $\eta^2 = .31$, and for group, Wilks's $\Lambda = .93$, $F(2, 342) = 21.39$, $p < .0001$, $\eta^2 = .07$. Students in third grade scored higher than those in second and first grades, and students not at risk for SMD scored higher than students at risk for SMD. All ANCOVAs were significant at the .001 level.

In summary, the results clearly show that all MANCOVAs were significant. Older children and children not at risk for SMD outperformed younger children and children at risk for SMD across aggregate measures of problem solving, arithmetic calculation, word-problem-solving processes, reading, phonological processes, STM, WM, random generation and updating, and vocabulary and semantic fluency. All effect sizes related to the MANCOVAs met or exceeded a Cohen's *d* of .50. Except for the calculation and reading measures, the effect sizes for the ANCOVAs as a function of ability group were low to moderate, suggesting that ability group differences are more reliable at the composite level.

Correlations

The next analyses examined the relationship between math problem solving and WM in the total sample. We predicted that if the WM system played an important role in accounting for age-related and individual differences in problem solving independent of the phonological system, then WM measures would predict problem-solving performance after various measures of the phonological system had been partialed from the analysis. We examined this hypothesis through a series of regression analyses in which phonological processes (i.e., STM, phonological knowledge) and WM were the independent variables and problem solving was the dependent measure. We also considered whether reading and calculation skill, semantic processing, speed, inhibition, and fluid intelligence mediated performance. Prior to our regression analysis, however, the intercorrelations between WM and problem-solving measures were examined.

Because all the MANCOVAs were significant, several of the above measures were aggregated into composite scores. This was done to cluster variables along theoretical lines, as well as for data reduction purposes. The composite scores were created by computing *z* scores for each task based on the total sample. The *z* scores for the appropriate tasks were then summed to create the composite scores. Two composite scores served as criterion measures: word problems (word-problem solving—semantic structure varied, WISC–III Arithmetic subtest, mean intercorrelation = .45), and mathematical computation (WRAT—arithmetic, WIAT—arithmetic, computation fluency, mean intercorrelation = .70). Three composite scores served as predictor variables that yielded one value for reading (WRAT—reading, WRT—comprehension, TOWRE—real words, mean intercorrelation = .88), phonological processing (elision, phonological fluency, TOWRE—pseudoword reading, mean intercorrelation = .45), and semantic processing (vocabulary, semantic fluency, $r = .47$). Because of the importance of speed in discussions of STM and WM (Kail & Hall, 2001; Salthouse, 1996), a composite measure of speed was also created (rapid naming of digits, rapid naming of letters, $r = .88$). In addition, because the correlation between the random-generation measures was weak ($r = .30$), the random generation of numbers task was used as the primary measure of inhibition.[2] In addition, the raw scores from the Raven Colored Progressive Matrices test served as a measure of fluid intelligence.

Although the composite scores fit a logical structure, we were uncertain about the structure of the memory measures. This was because several recent studies have argued that there are no pure measures of STM or WM and therefore that latent scores must be used (see, e.g., Conway, Cowan, Bunting, Therriault, & Minkoff, 2002; Engle et al., 1999). Thus, to reduce the data set, we submitted the memory measures to a principal factor analysis. Also included in the analysis was the updating measure. Because of the broad selection of measures and in consideration of our sample size, we relied on factor scores to keep the number of estimated parameters within a reasonable range. In addition, variance across the tasks was wanted variance, and task-specific variance was of less interest. As shown in Table 3, a two-factor model emerged (eigenvalues of 1.40 and 1.10, respectively). To interpret Table 3, we used a varimax rotation (orthogonal solution) and considered factor loadings of .30 as meaningful. We used the common factor analysis and varimax rotation because scores on each measure had a reasonable degree of reliability and shared common variance with scores on other measures. We were also interested in the independent contribution of each dimension in explaining the covariation of individual differences on the measures. Therefore, an orthogonal solution was used to retain the independent dimensions. As shown in Table 3, all STM measures (except backwards digit) and the updating task loaded on Factor 1. All WM tasks loaded on Factor 2.

To determine if the two-factor structure was an adequate extraction of the matrix, we obtained maximum-likelihood estimates (Jöreskog & Sörbom, 1984) for the two-factor model. The likelihood ratio chi-square test yielded $\chi^2(26, N = 353) = 32.84$, $p =$

[2] Measures of fluency are associated with inhibition in adult samples (see, e.g., Conway & Engle, 1994). However, the correlation between semantic fluency and phonological fluency, with age partialed out, in the present sample was weak ($r = .28$). A weak partial correlation also emerged between semantic fluency and random number generation ($r = -.16$) and between semantic fluency and random letter generation ($r = -.09$). Likewise, a weak partial correlation emerged between phonological fluency and random number generation ($r = -.08$) and between phonological fluency and random letter generation ($r = -.13$). Thus, the fluency measures did not appear to capture inhibition in the age groups represented here.

Table 3
Factor Structure of Short-Term Memory, Working Memory, and Updating Tasks

Task	1	2	3	4	5	6	7	8	9	10	Factors (varimax rotation) I	II
Short-term memory												
1. Digit forward	—										**.51**	−.001
2. Digit backward	.08	—									.24	**.36**
3. Pseudoword span	.30	.13	—								**.53**	.17
4. Real-word span	.30	.23	.38	—							**.55**	.16
Working memory												
5. Listening span	.11	.22	.19	.26	—						.24	**.40**
6. Digit/sentence	.13	.27	.21	.19	.29	—					.22	**.54**
7. Semantic association	.15	.11	.17	.15	.15	.27	—				.17	**.36**
8. Visual matrix	.08	.17	.09	.03	.19	.24	.10	—			.06	**.37**
9. Mapping/directions	−.05	.07	.10	.08	.15	.24	.22	.14	—		−.03	**.42**
Updating												
10. Updating	.35	.24	.37	.35	.19	.26	.14	.12	.01	—	**.57**	.18

Note. Boldfaced numbers in the factor columns indicate factor loadings > .35.

.16. Nonsignificance was considered one criterion for model acceptance discussed by Bentler and Bonett (1980). The goodness-of-fit index (Bentler & Bonett, 1980) was computed from the null model (which hypothesizes that the variables are uncorrelated) in the population, $\chi^2(45, N = 353) = 442.95$, $p < .001$, and the current two-factor model as (442.95 – 32.84/442.95) = .926. Thus, the model was 93% of the way to a perfect fit. Further testing of this model included an analysis of the χ^2/df ratio, the root-square residual, and the Tucker-Lewis Index (TLI). The χ^2/df ratio provided information on the relative efficiency of the alternative model in accounting for the data (Marsh, Balla, & McDonald, 1988). Values of 2.0 or less were interpreted as representing an adequate fit. The present two-factor model was 1.25. The root-mean-square residual (RMSR) measured average residual correlation (Jöreskog & Sörbom, 1984). Smaller values (e.g., .10 or less) reflected a better fit. The RMSR for the two-factor structure was .003. The TLI roughly scaled the chi-square from 0 to 1, with 0 representing the fit of the null model (Bentler & Bonett, 1980), which assumed that the variables were uncorrelated, and 1 represented the fit of a perfectly fitting model. Values less than .90 would suggest that the model could be improved substantially (see Marsh et al., 1988, p. 292, for discussion), whereas values close to 1.0 would indicate a better fit. This measure, when compared with the other indices, was relatively independent of sample size. The TLI in the present study was .97.

As shown in Table 4, the two factor scores (STM and WM) based on the common factor analysis were correlated with math, reading, fluid intelligence, and cognitive measures. Also included in the correlation analysis were chronological age and the problem-solving component score related to knowledge of algorithms. This latter measure was a subcomponent of the word-problem-processing task and was selected because it was found to eliminate the significant contribution of WM for samples in the higher grades (Swanson & Sachse-Lee, 2001). In addition, a factor analysis of component scores of the present data showed that the knowledge of algorithms loaded highest on the first factor. Thus, for simplicity, only the knowledge of algorithms component score was analyzed. As shown in Table 4, several significant correlations emerged. Because of the number of comparisons, alpha was set to .001 ($N = 353$, $rs > .18$, $p < .001$).

Table 4
Intercorrelations Among Mathematics, Reading, and Cognitive-Processing Variables

Variable	1	2	3	4	5	6	7	8	9	10	11	12
1. Age	—											
2. Word-problem accuracy	.50	—										
3. Arithmetic calculation	.79	.65	—									
4. Reading skills	.60	.72	.80	—								
5. Phonological knowledge	.41	.63	.62	.79	—							
6. Processing speed	−.49	−.58	−.63	−.72	−.60	—						
7. Semantic processing	.49	.62	.63	.67	.62	−.51	—					
8. Inhibition	−.19	−.31	−.25	−.34	−.28	.30	−.31	—				
9. Short-term memory	.25	.47	.40	.50	.48	−.44	.43	−.26	—			
10. Working memory	.37	.54	.51	.52	.49	−.37	.53	−.22	.26	—		
11. Algorithm knowledge	.36	.62	.42	.48	.41	−.33	.44	−.20	.32	.35	—	
12. Fluid intelligence	.43	.58	.55	.62	.51	−.42	.51	−.27	.37	.41	.42	—

Note. $rs > .21$, $p < .0001$.

An inspection of Table 4 shows four important findings (to interpret the results, we considered rs $> .50$ as substantial correlations). First, problem-solving accuracy was significantly related to all variables except inhibition ($p > .0001$). Second, phonological processes (phonological knowledge and STM) were significantly related to the majority of measures. Substantial correlations occurred between phonological processing and composite scores of word-problem-solving accuracy, arithmetic calculation, processing speed, and semantic processing. Third, WM was substantially correlated with arithmetic calculation, semantic processing, and fluid intelligence.

Finally, the correlation between WM and word-problem solving was substantial, $r(351) = .54$, $p < .0001$. To examine this relationship further, we partialed from the correlation analysis variables assumed to underlie individual differences in WM. The correlation between WM and problem solving was $r = .46$ when partialed for age, $r = .33$ when partialed for age plus arithmetic calculation, and $r = .30$ when partialed for age plus reading. When processing variables were considered, the partial correlation was $r = .32$ when partialed for age plus phonological processing, $r = .40$ when partialed for age plus speed, $r = .31$ when partialed for age plus semantic processing, $r = .44$ when partialed for age plus inhibition, $r = .42$ when partialed for age plus STM, $r = .41$ when partialed for age plus knowledge of algorithms, and $r = .37$ when partialed for age plus fluid intelligence. Thus, the smallest coefficient related to WM and problem solving partialed out the influence of age and reading. However, all the partial coefficients were significant (all ps $< .0001$), and none of the partial coefficients that included age differed significantly from the other partialed coefficients in magnitude (via Fisher z-score transformation, all ps $> .001$).

We analyzed the correlations between problem solving and WM within grades. A significant correlation between problem solving and WM was found in Grade 1, $r(128) = .57$, $p < .0001$; Grade 2, $r(88) = .42$, $p < .0001$; and Grade 3, $r(127) = .34$, $p < .0001$. We also partialed the influence of phonological processing, inhibition, and speed from these coefficients. The partial coefficients were $r = .46$, $p < .0001$; $r = .28$, *ns*; and $r = .25$, $p < .001$, for Grades 1, 2, and 3, respectively. These findings show that the magnitude of the zero-order correlations and partial correlations was greater in Grade 1 than in Grades 2 and 3. However, the magnitude of the coefficients was reduced when measures of processing efficiency (phonological processing, inhibition, and naming speed) were partialed from the analysis. It is important to note that the magnitude of the partial coefficients did not differ significantly between grades (ps $> .001$).

Predictions of Problem Solving

The next analysis determined those variables that significantly predicted word-problem solving when the effects of all other competing variables were partialed from the analysis. Subsequent regression analyses assessed the variables that contributed unique variance to problem-solving performance. We investigated whether the relationship between problem solving and WM was maintained when blocks of variables related to age, phonological processing, semantic processing, inhibition, and reading skill were entered into the analysis. We also addressed the same question when comparing the relationship between math calculation and WM. In summary, the regression analysis was done (a) to determine if the influence of WM was partialed out when variables related to phonological and related processes were entered into the analysis and (b) to determine if blocks of variables related to reading, math, inhibition, and speed increased the predictability of word-problem solving when compared with a model that included only WM.

The criterion and predictor variables were the same as those shown in Table 4. Criterion measures were z scores (converted from raw scores based on the total sample) from the problem-solving and math calculation measures. Predictor variables were z scores related to WM, STM phonological and semantic processing, reading, inhibition, chronological age, and the component score related to knowledge of algorithms.

For our first set of analyses, we determined the amount of variance in problem-solving performance that was accounted for by WM alone (Model 1). As shown in Table 5, WM contributed approximately 30% of the variance to problem solving and 26% of the variance to arithmetic calculation. For each subsequent model, variables were entered simultaneously such that the beta values reflected unique variance (the influence of all other variables partialed out). In Model 2, we determined the contribution of the factor scores related to STM, WM, and age when entered in the model. As shown, WM contributed significant variance to both math calculation and word-problem solving even when STM and age were partialed from the analysis. In addition, both STM and age contributed unique variance. The predictor variables in Model 2 contributed approximately 50% of the variance in word-problem solving and 62% of the variance in math calculation. When compared with Model 1, Model 2 significantly improved the predictions for math calculation, $F_{\text{inc}}(2, 345) = 163.42$, $p < .001$, and problem-solving accuracy, $F_{\text{inc}}(2, 345) = 69.03$, $p < .001$.

In Model 3, we assessed the contribution of WM to problem solving as well as when composite scores related to phonological processing, speed, and inhibition were added to Model 2. As shown, Model 3 accounted for 58% of the variance in word-problem solving and 70% of the variance in math calculation. All variables contributed significant variance to word-problem-solving accuracy. In contrast, all variables except STM and inhibition contributed unique variance to math calculation. Model 3 improved the prediction for calculation and problem solving when compared with Model 2, $F_{\text{inc}}(3, 342) = 30.57$, $p < .001$, and $F_{\text{inc}}(3, 342) = 22.16$, $p < .001$, respectively.

In Model 4, we determined whether the variables that contributed significant variance in Model 3 were eliminated when the reading and the semantic-processing composite scores were entered into the model. Model 4 captured approximately 61% of the variance in word-problem solving and 73% of the variance in math calculation. The important finding related to Model 4 is that the significant influence of phonological processing and processing speed was eliminated in the prediction of word-problem-solving accuracy. The only variables that contributed unique variance to solution accuracy were reading, semantic processing, age, STM, and WM. Likewise, Model 4 showed that reading, age, and WM contributed unique variance to math calculation. Model 4 improved the prediction for calculation and problem solving when compared with Model 3, $F_{\text{inc}}(2, 340) = 37.97$, $p < .001$, and $F_{\text{inc}}(2, 340) = 13.08$, $p < .001$, respectively.

Table 5
Hierarchical Analysis of Math Calculation and Word-Problem Solving

	Math calculation				Word-problem solving			
Model & variable	*B*	*SE*	β	*t* ratio	*B*	*SE*	β	*t* ratio
Model 1								
WM	1.89	.17	.51	11.04***	1.80	.14	.54	12.19***
Model 2								
Age	1.55	.09	.58	16.23***	0.73	.09	.31	7.50***
STM	0.64	.12	.19	5.34***	0.90	.12	.29	7.39***
WM	0.96	.12	.26	7.27**	1.18	.13	.36	8.72***
Model 3								
Inhibition	−0.13	.09	−.04	−1.42	−0.21	.09	−.08	−2.19*
Speed	−0.17	.05	−.12	−3.13***	−0.19	.05	−.15	−3.29**
Phonological processing	0.30	.04	.26	6.44***	0.24	.04	.24	4.88***
Age	1.25	.09	.46	13.29***	0.44	.09	.18	4.50***
STM	0.15	.13	.04	1.29	0.27	.14	.15	3.61***
WM	0.54	.12	.14	4.22**	0.77	.15	.25	6.00***
Model 4								
Reading	0.35	.05	.37	6.18***	0.26	.06	.31	4.28***
Semantic processing	0.10	.06	.06	1.57	0.15	.06	.11	2.29*
Inhibition	−0.07	.08	−.02	−0.81	−0.15	.09	−.06	−1.67
Speed	−0.03	.05	−.02	−0.55	−0.08	.06	−.06	−1.33
Phonological processing	0.09	.05	.08	1.70	0.06	.06	.06	1.13
Age	0.98	.09	.37	10.12***	0.21	.10	.09	2.09*
STM	0.02	.11	.01	0.20	0.33	.12	.11	2.71**
WM	0.37	.12	.10	2.94**	0.63	.19	.19	4.66***
Model 5								
Fluid intelligence	0.03	.09	.01	0.38	0.36	.10	.15	3.55**
Algorithm knowledge	0.26	.03	.39	7.93***	0.26	.06	.31	4.28***
Reading	0.34	.05	.36	5.86***	0.20	.06	.24	3.27**
Semantic processing	0.09	.06	.06	1.51	0.13	.06	.09	1.93
Inhibition	−0.06	.08	−.02	−0.77	−0.13	.09	−.05	−1.43
Speed	−0.03	.05	−.02	−0.58	−0.09	.06	−.08	−1.65
Phonological processing	0.09	.05	.08	1.61	0.06	.05	.06	1.13
Age	0.98	.09	.36	10.04***	0.18	.10	.07	1.79
STM	0.02	.11	−.01	0.17	0.30	.12	.05	2.48*
WM	0.37	.12	.07	2.89**	0.59	.13	.18	4.39***

Note. Model 1: Math calculation: $F(1, 347) = 121.32, p < .001, R^2 = .26$; Word-problem solving: $F(1, 347) = 148.65, p < .001, R^2 = .30$; Model 2: Math calculation: $F(3, 345) = 191.24, p < .001, R^2 = .62$; Word-problem solving: $F(3, 345) = 113.87, p < .001, R^2 = .50$; Model 3: Math calculation: $F(6, 342) = 131.31, p < .001, R^2 = .70$; Word-problem solving: $F(6, 342) = 77.45, p < .001, R^2 = .58$; Model 4: Math calculation: $F(8, 340) = 116.26, p < .001, R^2 = .73$; Word-problem solving: $F(8, 340) = 65.70, p < .001, R^2 = .61$; Model 5: Math calculation: $F(9, 339) = 103.10, p < .001, R^2 = .73$; Word-problem solving: $F(9, 339) = 61.79, p < .001, R^2 = .62$. WM = working memory; STM = short-term memory.
* $p < .05$. ** $p < .01$. *** $p < .001$.

In Model 5, we determined whether the variables that contributed significant variance in Model 4 were eliminated when measures of fluid intelligence were entered into the model. The important finding related to Model 5 is that the significant influence of age was eliminated in the prediction of word-problem-solving accuracy. The only variables that contributed unique variance to solution accuracy were fluid intelligence, reading, STM, and WM. In contrast, Model 5 provided a comparable pattern of results to Model 4 in predicting math calculation. The results showed that reading, age, and WM were the only variables that contributed unique variance to math calculation. Model 5 did not improve the prediction for calculation, $F < 1$, but did improve the prediction of problem-solving accuracy when compared with Model 4, $F_{\text{inc}}(2, 339) = 4.46, p < .01$.

A final regression was computed related to word-problem solving. For this model, we entered the composite score related to math calculation and the component score related to knowledge of algorithms into the regression model. The results are shown in Table 6. When comparing Model 5 in Table 5 with Model 6 in Table 6, one finds that R^2 increased by 7%. More importantly, the results show that WM remained a significant predictor of word-problem solving. The results also show that algorithmic knowledge and processing speed contributed unique variance to solution accuracy. Model 6 significantly improved the prediction for problem solving when compared with Model 5, $F_{\text{inc}}(3, 337) = 25.32, p < .001$.

In summary, there are two important findings related to the hierarchical regression analysis. First, in the complete model (see Table 6), only WM, STM, fluid intelligence, processing speed, and knowledge of algorithms contributed unique variance to problem-solving accuracy. Second, partialing out phonological processing, STM, speed, and inhibition did not eliminate the significant role

Table 6
Hierarchical Analysis for Cognitive Variables and Math Calculation Skill Predicting Word-Problem Solving

Variable	B	SE	β	t ratio
Model 6				
Fluid intelligence	0.25	.09	.11	2.62**
Algorithm knowledge	0.70	.08	.29	8.20***
Math calculation	0.09	.05	.11	1.79
Reading	0.12	.05	.14	2.03*
Semantic processing	0.05	.06	.04	0.91
Inhibition	−0.13	.08	−.08	−1.47
Speed	−0.11	.05	−.09	−2.05*
Phonological processing	0.05	.05	.05	1.03
Age	−0.01	.12	−.005	−0.01
STM	0.29	.11	.09	2.61**
WM	0.54	.13	.17	4.32***

Note. Model 6: $R^2 = .69$, $F(11, 337) = 67.18$, $p < .0001$. STM = short-term memory; WM = working memory.
* $p < .05$. ** $p < .01$. ** $p < .001$.

that WM plays in predicting problem solving. In fact, no significant variance could be attributed to phonological processing and inhibition when partialed for the influence of other variables.

DISCUSSION

The purpose of this study was to determine the mechanisms that mediate the relationship between WM and problem-solving accuracy in elementary school children. Two models were tested. Before discussing the results related to these models, however, we briefly summarize age-related and individual differences across the various problem-solving, achievement, and cognitive measures.

In terms of age-related differences, the results yielded the expected finding that older children outperform younger children on WM and problem-solving tasks. Although increases in WM and problem solving are age related, age accounted for only a small proportion of the variance in predicting problem-solving accuracy, suggesting that developmental or school-based processes other than increases in age mediated the relationship. That is, the correlation between WM and problem solving was moderate at $r = .54$ and was reduced to only $r = .46$ when partialed for chronological age. Furthermore, there was no significant difference in the magnitude of the correlation between WM and problem solving across the three grades. These results suggest that the influence of WM across grades is stable. In addition, in the full regression model, we found that chronological age did not contribute unique variance to problem solving.

In terms of ability group findings, performance of children at risk for SMD was below that of children not at risk on aggregate measures related to problem solving, calculation, reading, phonological processing, random generation/updating, STM, WM, and semantic processing/vocabulary. Of particular interest to our study was identifying those cognitive processes that, when partialed out of the analysis, would mitigate the relationship between individual differences in WM and problem-solving accuracy. The results show a significant relationship between WM and problem solving even when the influence of phonological processing, inhibition, speed, and math calculation and reading skill was partialed from the analysis. However, it is important to note in our study that none of the individual WM tasks separated the two ability groups at the .001 alpha level. This may be because WM and fluid intelligence share important variance (Engle et al., 1999; Kyllonen & Christal, 1990), and therefore, partialing out fluid intelligence in the analysis may have removed important variance related to ability group differences on individual tasks. Regardless, ability group differences in WM did emerge on aggregate WM scores, suggesting that latent measures that cut across the WM task better discriminate between ability groups than isolated tasks. This latter finding has emerged in other studies (e.g., Wilson & Swanson, 2001).

The important findings, however, relate to the two models of WM and its influence on mathematical word-problem solving. One model tests whether phonological processes (e.g., STM, phonological knowledge) play a major role in predicting performance in problem solving and whether the phonological system mediates the influence of executive processing (WM) on problem solving. Phonological processes in this study were related to factor scores of STM and composite scores of phonological knowledge (measures of elision, pseudoword reading, phonological fluency). The model follows logically from the reading literature that links phonological skills to new-word learning (see, e.g., Baddeley, Gathercole, & Papagno, 1998), comprehension (Perfetti, 1985), and mental calculation (see, e.g., Logie, Gilhooly, & Wynn, 1994). The model assumes that low-order processing, such as phonological coding, provides a more parsimonious explanation of ability group differences in problem solving than measures of WM do. Recall that in this model, WM regulates the flow of information from a phonological store. Thus, the model suggests that poor problem solvers have deficits in the processing of phonological information, creating a bottleneck in the flow of information to higher levels of processing.

The second model suggests that problem-solving performance relates to executive processing, independent of the influence of the phonological system. This assumption follows logically from the problem-solving literature suggesting that abstract thinking, such as comprehension and reasoning, requires the coordination of several basic processes (see, e.g., Engle et al., 1999; Just, Carpenter, & Keller, 1996; Kyllonen & Christal, 1990). Measures of executive processing in this study were related to a factor score of WM and measures assumed to reflect activities on the executive system, for example, inhibition (random generation of numbers) and activation of LTM (composite measures of reading, arithmetic calculation, knowledge of algorithms). The findings on these two models are as follows.

First, WM contributes unique variance to problem solving beyond what phonological processes (e.g., STM, phonological knowledge), as well as reading skill, calculation, inhibition, processing speed, and semantic processing, contribute. The results show that WM contributed approximately 30% variance to problem-solving accuracy when entered by itself in the regression analysis. Furthermore, although age-related and ability group–related differences emerged on measures of phonological processing, these measures did not partial out the influence of individual differences in WM on problem solving. There is clear evidence that the executive system of WM does contribute important variance to problem-solving performance beyond processes that relate to the phonological system. Thus, the results do not support the first model. There is weak support for the assumption that bottom-up processes (i.e., the phonological system) mediate individual differences in WM performance and its influence on solution accuracy. A refinement of this model may consider phonological processing as important to problem solving, but no more important than other processes.

No doubt, a theoretical problem emerges, however, when one considers how to reconcile specific phonological deficits (e.g., problems related to the phonological system) recently attributed to some individuals with SMD (see, e.g., Hecht, Torgesen, Wagner, & Rashotte, 2001) with the notion that an executive system may override such processes. One means of reconciling this issue with the current study would be to test (a) whether problems in specific activities of the central executive system can exist in individuals with SMD independent of their problems in phonological processing (Swanson, 1993) and/or (b) whether a general manipulation of processing and storage demands may indirectly account for low-order processing deficits (especially on language-related tasks). This issue can be put within the context of Baddeley's (1986, 1996; Baddeley & Logie, 1999) WM model. In Baddeley's (1986) model, the central executive system is an undifferentiated generic system that is used to support low-order systems. However, if the executive system is overtaxed, it cannot contribute resources to low-order processing. This is because the phonological loop is controlled by the central executive (Baddeley, 1996), and therefore, any deficits in phonological functioning may partially reflect deficiencies in the controlling functions of the central executive itself (see Baddeley, 1996; Gathercole & Baddeley, 1993).

Second, WM captures unique variance in predicting solution accuracy beyond measures of LTM and inhibition. Measures of LTM in this study were related to reading and calculation ability, as well as knowledge of algorithms. It has recently been argued that the executive system functions to access information from LTM (see, e.g., Baddeley & Logie, 1999). We found that the contribution of knowledge of algorithms to the regression equation contributed unique variance to solution accuracy. However, additional variables that uniquely predicted problem solving in the complete model were speed, STM, reading, and WM. This finding clearly supports the notion that components of executive processing (other than LTM, processing speed, and inhibition) contribute unique variance to solution accuracy. Thus, it seems there are activities besides the aforementioned variables that contribute unique variance to problem-solving ability. We posit that monitoring of processing and storage demands may be one aspect of this unique variance.

The study also addressed two hypotheses discussed in the literature as playing a major role in accounting for individual differences in WM. One hypothesis relates to the speed of processing. A simple version of this hypothesis states that individuals with SMD are slower at processing language information than are average-achieving children and that such reduced processing on the SMD participants' part underlies their poor WM performance. Several models of WM assume that operations related to language are time consuming (see, e.g., Salthouse, 1996). Therefore, speed of processing may underlie the general pattern of WM difficulties noted in the present study. Furthermore, Kail (1993) has argued that a common pool of cognitive resources related to processing speed is used to perform a variety of tasks, with the pool increasing across ages in children. Clearly, our findings show a significant relationship between processing speed and measures of problem solving, semantic processing, inhibition, STM, and WM, as well as reading and calculation (rs range from $-.33$ to $-.72$; see Table 4). However, when measures of fluid intelligence and reading were entered into the regression analysis in predicting problem solving and math calculation, speed did not contribute significant variance (see Model 5, Table 5). Although speed does play an important role in the complete model, it was only one of several variables that contributed unique variance. More importantly, the correlation coefficient between WM and problem solving was significant when partialing out the influence of speed and age.

A second hypothesis considers whether individuals with SMD are less resistant to interference (see Baddeley, 1996; Brainerd & Reyna, 1993; Rosen & Engle, 1997; Towse, 1998, for further discussion of this model). Such a hypothesis assumes that an inhibition deficit limits SMD participants' ability to prevent irrelevant information from entering WM during the processing of targeted information (see Passolunghi & Siegel, 2001, for discussion of this model). An activity related to the central executive that has been implicated as a deficit in children with reading disabilities is their ability to suppress irrelevant information under high processing demand conditions (see, e.g., Chiappe et al., 2000; De Beni et al., 1998; Swanson & Cochran, 1991). Earlier studies showed that children with reading disabilities vary from controls in their ability to recall targeted (relevant) and nontargeted (incidental) information (see, e.g., Swanson & Cochran, 1991). Likewise, individuals with SMD may have difficulty preventing unnecessary information from entering WM and, therefore, would be more likely to consider alternative interpretations of material (such as asked for in the processing questions) that are not central to the task when compared with average achievers. This interpretation fits within several recent models that explain individual differences in memory performance as related to inhibitory mechanisms (see, e.g., Cantor & Engle, 1993; Conway & Engle, 1994) without positing some form of a capacity deficit.

Our results show that children with SMD experienced difficulties on the number random-generation tasks when compared with children not at risk for SMD. However, partialing out performance on the inhibition measure did not eliminate the significant relationship between WM and problem solving. One could argue, however, that the process question in our WM tasks put demands on the participant's ability to suppress competing information and therefore that individual differences in inhibition mediated WM performance. Clearly, the process questions for the current WM tasks constituted a temporary competing condition with storage. As a consequence, children at risk for SMD may have had difficulty preventing unnecessary information from entering WM and, therefore, considered alternative interpretations of material (such as those asked for in the process questions) that were not central to the task. Although we see the above model as a viable alternative to the results, we have three reservations. First, only the span levels of participants who answered the process question correctly were analyzed. If a process question was missed, the participants' recall of previously stored information was not scored. This procedure is different from previous studies (e.g., Daneman & Carpenter, 1980) that have allowed dissociation between the process question (i.e., it is not necessary for participants to answer the process question correctly) and the retrieval question in the analyses. Our procedure removed from the analysis irrelevant responses that emerged between the processing of the distractor question and the retrieval question. Second, if children at risk for SMD suffer more interference (i.e., diminished inhibition in that a large number of traces are simultaneously active) than children not at risk for SMD, then one would expect the effects of WM to be nonsignificant in predicting both word-problem solving and calculation when measures of inhibition are partialed from the analysis. Such was not the case in this study. That said, we do assume that inhibition efficiency may be a consequence of capacity constraints (see Cantor & Engle, 1993, for discussion). Children at risk for SMD may use more WM capacity than children not at risk for SMD to inhibit or resist potential interference from irrelevant items (see Chiappe et al., 2000, for discussion).

Overall, these findings give partial support to Model 2. We say partial because the role of the executive system has not been clearly delineated. We argue that because partialing out speed, inhibition, and general knowledge (e.g., arithmetic calculation, knowledge of algorithms) did not eliminate the significant effects of WM on problem solving, individual differences in WM are related to constraints in regulating and/or manipulating storage capacity. Of course, this study provides no direct measure of mentally coordinating resources across the WM tasks, and therefore, one could argue that the results merely reflect the fact that WM tasks draw on finite resources from a phonological storage system. We argue, however, that the executive system coordinates the distribution of finite resources of the verbal and visual–spatial system. Support for this common system is found in the factor analysis in that both verbal and visual–spatial WM loaded on the same factor (see Table 3), whereas STM tasks loaded on a separate factor.

Implications

There are three implications of our findings for current literature. First, bottom-up processes (e.g., the phonological system) are not the primary mediators between age-related and individual differences in WM and problem solving. Of course, these results apply only to the age and ability groups represented in this sample. However, similar results have occurred with older children (Swanson, in press; Swanson & Sachse-Lee, 2001). Our findings further suggest that although skills associated with phonological processes (i.e., naming speed and STM) are important to age-related changes in children in calculation and problem solving, they are no more important than WM. Such a finding qualifies bottom-up models of problem solving of children by suggesting that if low-order processes, such as phonological processes (e.g., STM), moderate the influence of executive processing (WM) on problem-solving performance, their effects may be indirect or minimal for children who have perhaps met a minimum threshold in mathematics and reading skills.

The second implication relates to the independence of WM and STM. We argue that STM and WM may make independent contributions to problem solving because STM measures draw on phonological codes (Salame & Baddeley, 1982), whereas WM measures draw on resources from the executive system (see, e.g., Engle et al., 1999). Our results are consistent with those of others who have argued that STM tasks and WM tasks are inherently different (Engle et al., 1999). We found separate loadings for STM and WM factors. That is, although phonological coding might be important to recall in STM, it may not be a critical factor in WM tasks. This finding is important because a common opinion is that STM tasks are a proper subset of processes of which WM is capable. As a qualification to this view, however, the present analysis shows that components of WM operate independent of STM. This finding is consistent with other experimental work with adults (see, e.g., Engle et al., 1999) and poor readers (Swanson et al., 1996). The implication of this finding is that problems in WM may co-occur with STM but also maintain some independence from the development of STM.

Some comment is necessary as to why children at risk for SMD suffer deficits on aggregate scores related to both STM and WM tasks. The above research suggests that children with SMD perform poorly on tasks that require accurate recognition/recall of letter and number strings or real words and pseudowords. Tasks such as these, which have a "read in and read out" quality to them (i.e., place few demands on LTM to infer or transform the information), reflect STM. One common link among these tasks is the ability to store and/or access the sound structure of language (phonological processing). There is evidence that participants with SMD suffer deficits in STM, a substrate of the phonological system. However, some children with SMD also do poorly on tasks that place demands on attentional capacity, a characteristic of WM tasks. The findings on STM tasks make sense to us because there are many mnemonic situations in which a stimulus in memory is attended to and the other stimuli exist as a background—that is, they are not the center of current awareness. These situations, in our opinion, do not challenge monitoring. We argue that WM tasks require the active monitoring of events and that these events are distinguishable from simple attention to stimuli held in STM. Monitoring within WM implies attention to the stimulus that is currently under consideration together with active consideration (i.e., attention) of several other stimuli whose current status is essential for the decision to be made. Results from our lab have suggested that the tasks differ in subtle ways. Although a substrate of STM may contribute to problems in verbal WM, children with SMD can suffer problems in a substrate of WM that are independent of problems in verbal STM. Problems in verbal WM have been found to persist in children with average IQ and learning problems even after partialing out the influence of verbal STM (Swanson et al., 1996), verbal articulation speed (Swanson & Ashbaker, 2000), reading comprehension (Swanson, 1999), or fluid intelligence (Swanson & Sachse-Lee, 2001).

Finally, although the influence of individual differences in WM on problem-solving performance is robust, this does not mean that its influence cannot be compensated for. As previously stated, increased performance on measures related to speed and knowledge of algorithms can reduce the influence of individual differences in WM on problem solving. What remains to be studied is the influence of instruction and age-related development on these processes.

Conclusion

In summary, our findings converge with studies on individual differences that suggest WM plays a critical role in integrating information during problem solving. We argue that WM plays a major role because (a) it holds recently processed information to make connections to the latest input and (b) it maintains the gist of information for the construction of an overall representation of the problem. Yet WM is not the exclusive contributor to variance in problem-solving ability. This study also supports previous research about the importance of reading skill, processing speed, and accessing information from LTM (knowledge of algorithms) in solution accuracy. Moreover, our findings are consistent with models of high-order processing suggesting that WM resources activate relevant knowledge from LTM (Baddeley & Logie, 1999; Ericsson & Kintsch, 1995) but also suggesting that a subsystem that controls and regulates the cognitive system plays a major role (Baddeley, 1986). Thus, we think one of the core problems children face in solving mathematical word problems relates to operations ascribed to a central executive.

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